

NOVELLIA PHARMACEUTICALS · FIELD FORCE TRAINING SERIES ·

FICTIONAL PRODUCT — EDUCATIONAL USE ONLY

Cardiovex® (Cardiovexaban) for

Secondary Prevention of MACCE

Level 1 Socratic Training Simulation — Annotated Transcript

Simulated Physician	Dr. Patricia Holloway, PharmD — Director of Pharmacy & P&T Committee Chair
Drug & Dose	Cardiovex® (cardiovexaban) 150 mg once daily PO Dual P2Y12/PAR-1 Antagonist
Company	Novellia Pharmaceuticals (<i>fictional entity</i> — <i>training purposes only</i>)
Difficulty	Level 1 of 3 Foundational Clinical Proficiency
Therapeutic Area	Cardiology / Secondary Prevention · Novel Antiplatelet Agent
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This annotated transcript captures a representative's call with Dr. Patricia Holloway, a seasoned P&T committee chair at a large health system. Unlike a passive inservice, Dr. Holloway pushes back, demands specifics, and rewards intellectual honesty over promotional polish. This is Level 1 — she wants depth, but she's not trying to trap you. She's evaluating whether you understand the drug well enough to be useful. Annotations appear after each exchange to unpack the clinical

concepts, explain the strategy behind strong answers, and flag the landmines to avoid.

Skills Being Tested

Approved Indication & Patient Selection	Can you define who qualifies — and clearly exclude who doesn't?
Biomarker & Risk Score Threshold	ASCVD risk calculation, clinical interpretation, and edge cases
SHIELD-1 & SHIELD-2 Trial Data	Efficacy endpoints and how to communicate them accurately
Statistical vs. Clinical Meaningfulness	The NNT concept applied to MACCE reduction
Intellectual Honesty Under Questioning	Naming limitations before they're found; not overselling

How to Use This Document: Read each exchange as it unfolds. Before reading the annotation, pause and ask yourself — what was the clinical concept at stake? What would you have said? The annotation is the learning layer. The dialogue is the test.

Exchange 1 of 5 — Opening: Patient Profile

Dr. Patricia Holloway leans back in her chair and sets down her coffee mug.

She gestures toward the chair across from her desk.

PATRICIA:

"Thanks for making the trip over. I know you've been trying to get on my calendar for a few weeks now."

She picks up her tablet and swipes to what looks like a clinical summary.

PATRICIA:

"So here's where I'm at — I've got two cardiologists who've been emailing me SHIELD-1 abstracts and asking when we're adding Cardiovox to formulary for secondary prevention. And honestly, I'm intrigued. We've had the usual suspects — aspirin, clopidogrel, ticagrelor — for years. But they keep telling me this is different because of the dual mechanism."

She looks up at you directly.

PATRICIA:

"Before I start thinking about P&T committee presentations or budget impact, help me understand this from the ground up. Which cardiovascular patients are we actually talking about here? Because I'm guessing this isn't for my 55-year-old with well-controlled hypertension and an LDL of 70 on a statin who's never had an event."

She waits, pen ready to take notes.

PATRICIA:

"Walk me through the patient profile."

YOU:

"You're right, it's a specific population. We're talking about adult patients with documented atherosclerotic cardiovascular disease — that means prior MI, prior ischemic stroke, or established peripheral arterial disease. These are secondary prevention patients who've already had an event.

The other key criterion is elevated residual risk despite guideline-directed medical therapy. Specifically, they need to have an ASCVD 10-year risk score of 15% or higher, or at least two additional high-risk features: diabetes with target organ damage, chronic kidney disease stage 3 or higher, or recurrent events within the past 24 months.

So the short version: it's your post-MI, post-stroke, or PAD patients who are already on aspirin and a statin, already optimized on blood pressure meds, but still carrying significant residual cardiovascular risk. Not every CV patient —

probably 30 to 40 percent of your secondary prevention population qualifies based on that risk threshold."

CLINICAL ANNOTATION — Exchange 1

What Just Happened — and Why It Worked

WHAT PATRICIA IS REALLY ASKING: She's not asking you to read the label back to her. She's asking you to demonstrate that you understand WHO this drug is for — and more importantly, who it is NOT for. Her example of "the 55-year-old with well-controlled hypertension" is a deliberate test: she's describing a primary prevention patient at low risk, and asking you to confirm Cardiovox isn't for them. If you hedge or say "well, it could apply to broader CV patients," you've failed.

WHY THE ANSWER WORKED:

- Led with documented ASCVD as the gating requirement — prior MI, stroke, or PAD. This matters because it signals you understand Cardiovox is for secondary prevention in patients who have already demonstrated atherosclerotic disease, not primary prevention.
- Named the risk stratification criteria precisely: ASCVD risk $\geq 15\%$ OR ≥ 2 high-risk features (diabetes with target organ damage, CKD stage 3+, recurrent events within 24 months). These are the exact enrollment criteria from SHIELD-1 and SHIELD-2, and they're on-label requirements.
- Connected the eligibility to residual risk despite GDMT: this is more than checking boxes — it's a risk phenotyping tool. The patient is already optimized on standard therapy but still at elevated risk.
- Quantified the target population without overstating it: "30 to 40 percent." This manages expectations and signals market awareness.

KEY CLINICAL CONCEPT — Residual Cardiovascular Risk:

Even with optimal statin therapy (LDL < 70 mg/dL), antiplatelet agents (aspirin or P2Y₁₂ inhibitor), and blood pressure control, many post-MI or post-stroke patients

retain elevated risk for recurrent MACCE (Major Adverse Cardiovascular and Cerebrovascular Events). This "residual risk" is driven by inflammatory pathways, platelet reactivity variability, and non-LDL lipid abnormalities that standard therapy doesn't fully address. Cardiovox targets two distinct platelet activation pathways simultaneously (P2Y12 and PAR-1), providing deeper platelet inhibition than single-pathway agents.

KEY CLINICAL CONCEPT — ASCVD Risk Score:

The ASCVD (Atherosclerotic Cardiovascular Disease) risk calculator estimates 10-year risk of heart attack or stroke based on age, sex, race, cholesterol levels, blood pressure, diabetes status, and smoking. A score of $\geq 15\%$ represents elevated residual risk in a secondary prevention population already on guideline-directed therapy. Scores below 15% suggest lower risk — not necessarily candidates for intensified antiplatelet therapy given bleeding risk.

KEY CLINICAL CONCEPT — Guideline-Directed Medical Therapy (GDMT):

The approved indication requires patients to be on GDMT before adding Cardiovox. "Guideline-directed medical therapy" means the patient is already on:

- **Antiplatelet agent** — aspirin 81 mg or clopidogrel
- **Statin** — high-intensity statin targeting LDL $<70\text{ mg/dL}$
- **Antihypertensive therapy** — if indicated, targeting BP $<130/80$
- **Diabetes management** — if diabetic, HbA1c $<7\%$

A patient optimized on GDMT who maintains elevated ASCVD risk is the target — not someone whose risk factors are uncontrolled.

⚠ WHAT TO WATCH FOR: Never say "Cardiovox can be used across cardiovascular patients" or imply it's appropriate for primary prevention. Patricia knows this — and payers will enforce secondary prevention + risk criteria at prior authorization.

Exchange 2 of 5 — ASCVD Risk Threshold: Pushing on the Cutoff

Dr. Holloway nods slowly and jots down "≥15% ASCVD risk OR ≥2 high-risk features, on GDMT, documented ASCVD" on her tablet.

PATRICIA:

"Okay, that's helpful. So we're talking about a true high-risk secondary prevention population — patients who should have lower risk on standard therapy but don't."

She taps her pen on the desk.

PATRICIA:

"Let me push on the ASCVD threshold for a second, because this matters for how we write the criteria. You said 15% or higher. Is that a hard floor from the trial design, or is there data below that cutoff? Because I've got patients at 12%, 13% — close but not quite there — and if one of my cardiologists makes the case that they're clinically similar, I need to know if there's any evidence to support going off-label below 15%."

She looks at you expectantly.

PATRICIA:

"What happens to efficacy below that threshold? Did they look at subgroups, or is 15% just where the line was drawn?"

YOU:

"That's the right question to push on, and I want to be straight with you: 15% is a hard enrollment floor from the trial design, not a threshold that emerged from subgroup analysis showing a cliff effect below it.

Both SHIELD-1 and SHIELD-2 required ASCVD risk scores of at least 15% at screening to enroll, unless patients met the alternative high-risk feature criteria. Patients below 15% without qualifying high-risk features were excluded — so we simply don't have randomized data below that cutoff. There's no subgroup analysis showing what happens at 12% or 13% because those patients weren't in the trials.

What I can tell you is that the FDA indication reflects exactly that population — patients with elevated residual risk as defined by that 15% threshold or the high-risk feature set. So for your patients at 12% or 13% without other qualifying features, there's no label support and no clinical trial data. That's not me being evasive — it's the honest answer, and I think you'd want me to tell you that rather than imply there's flexibility that isn't there.

The one nuance worth noting: ASCVD risk scores fluctuate based on lipid levels, blood pressure control, and changes in diabetes status. A patient at 13% today

who develops worsening renal function or has a second minor event could cross into the qualifying range. So the 15% threshold is about sustained elevated risk, not a single point-in-time calculation."

CLINICAL ANNOTATION — Exchange 2

The Directness Test — When Intellectual Honesty Is the Strategy

WHAT PATRICIA IS REALLY ASKING: She's testing whether you'll get creative with data that doesn't exist, or whether you'll defend the label honestly when a physician pressure-tests it. The 12%/13% question is a real-world scenario that happens constantly — a cardiologist sees a patient they believe would benefit and starts lobbying for off-label use. Patricia needs to know whether to support that or hold the line for PA purposes.

WHY THE ANSWER WORKED:

- Immediately named the limitation: "we don't have randomized data below that cutoff." No hedging, no "well, some clinicians think..." Just the truth.
- Distinguished between a design floor and an evidence-based threshold — this is a subtle but important distinction. The 15% cutoff isn't a cliff where efficacy falls off — it's simply where enrollment was set. That matters for how Patricia frames it to her medical staff: "We don't know it doesn't work below 15% — we just don't have data either way."
- Connected it to FDA labeling, which gives Patricia the regulatory grounding she needs for P&T documentation.
- Ended with the risk score variability nuance — this is clinically sophisticated and shows the rep understands risk stratification vs. a point-in-time calculation.

KEY CLINICAL CONCEPT — ASCVD Risk Score Variability:

ASCVD risk scores change based on evolving clinical parameters: LDL levels (if statin dose is adjusted), blood pressure (if hypertension worsens or improves), smoking status, development of new diabetes, or progression of existing diabetes.

A patient whose score is 13% today may have been 17% six months ago or may rise above 15% as renal function declines. The trial design used the screening risk score (prior to randomization) as the qualifying threshold — not a single clinic value.

KEY CLINICAL CONCEPT — Hard Enrollment Floor vs. Evidence-Based Threshold:

These are different things. A *hard floor* means the trial excluded patients below it — full stop. An *evidence-based threshold* means data exist showing that efficacy changes below the cutoff. SHIELD-1 and SHIELD-2 set 15% as an enrollment floor. There is no subgroup showing a cliff at 14% — but there's also no data showing the drug works below 15%. The honest answer acknowledges both.

PA IMPLICATIONS FOR FORMULARY:

Most commercial and PBM prior authorization criteria will require documented ASCVD risk $\geq 15\%$ or qualifying high-risk features. No trial data means no clinical justification for PA exceptions below that threshold. Patricia flagged this herself — it's exactly how payers will implement this.

⚠ **WHAT TO WATCH FOR:** Do NOT say something like "well, there's biological plausibility for patients at 13%." That's speculating beyond the data, and a P&T pharmacist will call it out immediately. Stick to what you know and what the label says.

Exchange 3 of 5 — MACCE Efficacy: SHIELD-1 and SHIELD-2

Dr. Holloway stops writing and looks up, clearly appreciating the directness.

PATRICIA:

"Good. That's exactly the kind of answer I need. No wiggle room for off-label creativity below 15% — I can work with that. And it makes sense from a payer perspective too. They're going to want documentation of that 15% threshold for PA, and now I know there's no trial data to argue against it."

She makes a note: "No data

<

15% risk — hard PA requirement."

PATRICIA:

"The variability piece is interesting though. So if I've got a patient who qualifies at 17%, starts therapy, and then their LDL improves and their calculated risk dips to 13% on treatment — that's not a reason to stop, right? We're qualifying them based on their baseline residual risk profile, not chasing the score once they're on drug."

She sets down her pen and leans forward slightly.

PATRICIA:

"Okay, so let's talk about what you're actually buying with this drug in that population. You mentioned SHIELD-1 and SHIELD-2 — give me the MACCE data. What's the hazard ratio, and more importantly, what does that actually mean for a patient who's already on aspirin and a statin and still at 15 to 20 percent 10-year risk?"

She watches you carefully, clearly expecting specifics.

YOU:

"You're right on the risk score question — the label and trial design support qualifying on baseline risk phenotype, not tracking the calculator on treatment. Once they're in, you're treating the underlying thrombotic risk, not the ASCVD score itself.

On the MACCE data — SHIELD-1 is the primary trial, SHIELD-2 confirmed it. In SHIELD-1, the primary endpoint was time to first MACCE — cardiovascular death, MI, stroke, or urgent coronary revascularization. Hazard ratio 0.75, 95% CI 0.64–0.88, $p < 0.001$. SHIELD-2 was consistent — hazard ratio 0.73, same four-point MACCE composite.

But you asked the right follow-up, so let me translate that. A hazard ratio of 0.75 means a 25% relative risk reduction. For your patient with a 15% 10-year ASCVD risk on GDMT, that translates to roughly a 3 to 4 percentage point absolute risk reduction over the trial duration. Expressed as number needed to treat: you'd need to treat approximately 33 to 40 patients for three years to prevent one MACCE event.

That's not a rounding error. Each one of those MACCE events is either a fatal MI, a stroke with permanent disability, or an urgent revascularization procedure. For a population that's already optimized on standard therapy and still carrying that risk, the marginal benefit is meaningful — especially when you factor in the downstream costs of recurrent events, rehospitalizations, and loss of functional independence."

CLINICAL ANNOTATION — Exchange 3

Numbers Are Just Numbers — Until You Make Them Real

WHAT PATRICIA IS REALLY ASKING: Two things at once. First, she's confirming the on-treatment monitoring principle (don't stop the drug because the risk score improves). Second, she's asking for the actual efficacy data AND asking you to contextualize it. The "what does that actually mean for a patient" language is a tell — she already knows the hazard ratios exist. She wants to see if you can translate statistics into clinical reality.

WHY THE ANSWER WORKED:

- Confirmed the on-treatment monitoring question cleanly and immediately — didn't linger, moved on. This is pacing: give a crisp answer and advance.
- Led with the precise numbers: HR 0.75 (SHIELD-1), HR 0.73 (SHIELD-2). Confidence interval and p-value included — these are what P&T wants to see in a slide deck, and reciting them signals mastery.
- Immediately translated the hazard ratio into plain language: "25% relative risk reduction" and then into absolute risk terms: "3 to 4 percentage point absolute reduction."
- Converted to NNT (Number Needed to Treat): "33 to 40 patients for three years to prevent one MACCE." This is the metric P&T committees use for budget impact modeling.
- Reframed the question from "is 1 fewer event per 35 patients meaningful?" to

"what is the cost of each MACCE event you're preventing?" This is the correct lens for a health system pharmacist focused on total cost of care.

KEY CLINICAL CONCEPT — Hazard Ratio (HR) vs. Absolute Risk Reduction (ARR):

A hazard ratio of 0.75 means the cardiovascular group had 25% fewer MACCE events than placebo over the trial duration. But the absolute difference depends on the baseline event rate. In SHIELD-1, the 3-year MACCE rate was approximately 12% in the cardiovascular arm vs. 16% in placebo — a 4 percentage point absolute difference. For a patient with 15% baseline 10-year risk, the absolute benefit is slightly lower (~3.5 percentage points).

KEY CLINICAL CONCEPT — Number Needed to Treat (NNT):

NNT is the number of patients you need to treat to prevent one adverse outcome. It's calculated as $1 / ARR$ (absolute risk reduction). An NNT of 33 means treating 33 patients for the trial duration (3 years) prevents 1 MACCE event. Lower NNT = more efficient therapy. For context:

- High-intensity statins for secondary prevention: NNT ~30–40 for CV events
- Aspirin for secondary prevention: NNT ~50–60 for CV events
- Cardioex (cardioexaban): NNT ~33–40 for MACCE

KEY CLINICAL CONCEPT — What a MACCE Event Actually Costs:

Each MACCE event includes:

- **Cardiovascular death** — terminal outcome
- **MI (myocardial infarction)** — average hospitalization cost \$15,000–\$30,000+, rehab, loss of work productivity
- **Stroke** — average cost \$50,000–\$100,000+ for acute care, long-term disability, rehospitalizations
- **Urgent coronary revascularization** — PCI or CABG, \$20,000–\$50,000+ per procedure

Beyond direct costs, recurrent events drive higher long-term healthcare utilization, functional decline, and reduced quality of life.

KEY CLINICAL CONCEPT — SHIELD-1 vs. SHIELD-2:

Both were Phase 3 randomized controlled trials of cardiovexaban 150 mg daily vs. placebo in high-risk secondary prevention patients on GDMT. SHIELD-1 (N=8,200) was the primary registration trial. SHIELD-2 (N=7,800) was the confirmatory trial. Both ran for 36 months (3 years). The consistency of findings across two independent RCTs — HR 0.75 and 0.73 — is a meaningful statement about reproducibility.

⚠ **WHAT TO WATCH FOR:** Hazard ratios can be misread. "0.75" doesn't mean 75% efficacy — it means the event rate in the treatment arm was 75% of the placebo rate (i.e., 25% lower). Some reps accidentally say "patients had a 75% reduction" — that's wrong and a P&T pharmacist will catch it immediately.

Exchange 4 of 5 — Disaggregating the Composite: CV Death and Stroke
Specifically

*Dr. Holloway nods, writing down "HR 0.75 SHIELD-1, 0.73 SHIELD-2 —
~25% RRR, NNT ~35."*

PATRICIA:

"Okay, I'll give you that. A 25% relative reduction with an NNT of 35 in a population already optimized on standard therapy is clinically meaningful — especially if we're preventing the hard endpoints like death and stroke. Those are the ones that drive long-term disability and mortality."

She pauses, tapping her pen against the tablet.

PATRICIA:

"But here's where I need you to be precise with me, because this is going to come up in P&T. You said MACCE — that's a four-point composite endpoint: CV death, MI, stroke, urgent revascularization. Break that down for me. What percentage of the benefit was driven by preventing hard outcomes like death and stroke versus softer endpoints like urgent revascularization? Because preventing a revascularization is nice, but what I really care about — what my CFO really cares about — is keeping people alive and out of the stroke unit."

She looks at you directly.

PATRICIA:

"Did they report the hard endpoint components separately, or is it all rolled into that composite? And if they did break it out, what did it look like?"

YOU:

"They did break it out, and the hard endpoint data is actually one of the stronger signals in SHIELD-2 specifically.

In SHIELD-2, the three-point MACE endpoint — CV death, MI, and stroke, excluding urgent revascularization — showed a hazard ratio of 0.71, 95% CI 0.59–0.85. That's a 29% relative risk reduction on the outcomes that matter most for long-term survival and disability.

For CV death alone, the hazard ratio was 0.78 — directionally favorable but with wider confidence intervals given the smaller number of events. For stroke specifically, the hazard ratio was 0.68 — a 32% relative risk reduction, and statistically significant.

SHIELD-1 reported similar directional findings on the hard endpoints, though the primary analysis was powered around the four-point MACCE composite.

I'll be honest with you about the limitation here: CV deaths were relatively infrequent in both trials — partly because these were well-managed secondary prevention patients on GDMT, so baseline mortality was lower than you'd see in older, less-optimized populations. The authors acknowledged that in the SHIELD-2 discussion. So the CV death subgroup is smaller, the confidence intervals are wider, and you shouldn't lead with that number the way you'd lead with the composite or the three-point MACE.

But the direction is consistent across all components, the stroke reduction is robust, and for a population that's already optimized on standard therapy, the marginal benefit on hard outcomes is what separates this from just another incremental antiplatelet agent."

CLINICAL ANNOTATION — Exchange 4

Composite Endpoints, Hard vs. Soft Outcomes, and the Art of the Honest Caveat

WHAT PATRICIA IS REALLY ASKING: She's disaggregating the endpoint — a standard P&T move. Composite endpoints (CV death OR MI OR stroke OR revascularization) are often used in clinical trials because they increase event rates and power the study. But composite endpoints can obscure what's driving the result. Patricia's CFO question is the right economic lens: death and stroke are where the clinical and economic value lives.

WHY THE ANSWER WORKED:

- Led with yes — they did break it out. Never make a P&T pharmacist wonder if the data exists.
- Gave the specific SHIELD-2 numbers: three-point MACE HR **0.71**, stroke HR **0.68**, CV death HR **0.78**. These are usable in a P&T slide.
- Immediately acknowledged the limitation without being asked: CV death events were infrequent, smaller subgroup, wider confidence intervals. This is advanced credibility-building — you surface the weakness before they can find it.
- Held the line on the overall conclusion: direction is consistent, stroke reduction is robust, and the hard endpoint benefit is what differentiates cardiovascular from incremental improvements.

KEY CLINICAL CONCEPT — Composite Endpoints vs. Hard Outcomes:

Clinical trials use composite endpoints (multiple events grouped together) to power the study — more events means more statistical power. But composites don't reveal which component is driving the benefit. In cardiovascular trials:

- **Hard outcomes:** CV death, stroke (permanent disability, high mortality risk)
- **Softer outcomes:** urgent revascularization (procedural intervention, lower mortality impact)

When Patricia says "break it down," she's asking whether the composite result is driven by preventing many revascularizations (less clinically meaningful long-term)

or by preventing some deaths and strokes (highly meaningful).

KEY CLINICAL CONCEPT — Three-Point MACE vs. Four-Point MACCE:

- **Three-point MACE:** CV death, MI, stroke (hard endpoints only)
- **Four-point MACCE:** CV death, MI, stroke, urgent coronary revascularization (adds softer endpoint)

Three-point MACE is considered the "harder" endpoint set because it excludes procedural interventions and focuses on irreversible outcomes.

KEY CLINICAL CONCEPT — Hazard Ratio Interpretation for Components:

A hazard ratio of 0.68 for stroke means the risk of experiencing a stroke at any given point in time is approximately 32% lower with cardiovexaban compared to placebo. For a CFO conversation, this translates to: "In a population of 1,000 high-risk patients treated for 3 years, you'd prevent approximately 12 strokes."

⚠ **WHAT TO WATCH FOR:** Don't lead with the CV death subgroup numbers as if they're the primary endpoint — they're not, and a P&T reviewer will note that. Use them as supporting evidence for the composite result, not as the headline.

Exchange 5 of 5 — Bleeding Risk: The Other Side of the Equation

Dr. Holloway writes down "3-pt MACE: HR 0.71, Stroke: HR 0.68, CV death:

HR 0.78" and puts an asterisk next to it.

PATRICIA:

"Okay, so the hard endpoint signal is there, and the stroke reduction is particularly compelling. That's the outcome that drives long-term disability costs and loss of independence."

She looks back at her notes.

PATRICIA:

"But here's the thing. When I take this to P&T, I'm going to have hospitalists and cardiologists who are going to say, 'Great, we're preventing MACCEs — but what about bleeding?' Because this is a dual-pathway antiplatelet agent, and we know that deeper platelet inhibition comes with bleeding risk. You can't give me efficacy data without the safety context."

She leans back in her chair.

PATRICIA:

"So talk to me about major bleeding. TIMI major bleeding, or whatever scale they used. What was the absolute increase, and how does it compare to the MACCE benefit we just discussed? Because if I'm preventing one stroke but causing one major GI bleed, that's not a favorable trade."

She looks at you directly.

PATRICIA:

"Give me the bleeding numbers, and then tell me how you'd frame the benefit-risk balance for a P&T committee."

YOU:

"I'm going to give you the numbers straight, because you're asking exactly the right question.

In SHIELD-1, the rate of TIMI major bleeding was 3.8% in the cardiovascular arm versus 2.1% in placebo over 3 years. Hazard ratio 1.82, statistically significant. SHIELD-2 showed similar findings: 4.1% versus 2.3%, hazard ratio 1.77.

So yes, there's an absolute increase of approximately 1.7 to 1.8 percentage points in major bleeding over 3 years. That's real, and I'm not going to minimize it.

Now let me frame the benefit-risk trade. The absolute MACCE reduction was approximately 4 percentage points. The absolute major bleeding increase was

approximately 1.8 percentage points. So for every 100 patients treated for 3 years, you prevent roughly 4 MACCE events and cause roughly 2 major bleeding events.

The nuance that matters for P&T: most of the bleeding events were GI bleeds — approximately 70% of the major bleeding events. Very few were intracranial hemorrhages, which are the catastrophic bleeds. ICH rates were low and not significantly different between groups: 0.4% cardiovexaban vs. 0.3% placebo.

So the honest framing for your committee: this is a dual-pathway antiplatelet agent with meaningfully deeper platelet inhibition than single-agent therapy. That deeper inhibition prevents hard cardiovascular outcomes — especially stroke — but it comes with increased GI bleeding risk. The trade is approximately 2 MACCE events prevented for every 1 major bleeding event caused.

For your high-risk secondary prevention patients — post-MI, post-stroke, PAD with recurrent events — that's likely a favorable trade. For lower-risk patients or patients with prior GI bleeding, it's not. That's why the risk stratification criteria matter. You're selecting the population where the benefit outweighs the harm."

CLINICAL ANNOTATION — Exchange 5

The Benefit-Risk Conversation — Where Credibility Is Won or Lost

WHAT PATRICIA IS REALLY ASKING: This is the test of whether you understand that efficacy data in isolation is meaningless. Every drug has a benefit-risk profile, and antiplatelet agents have a well-known bleeding liability. Patricia is asking you to present both sides of the equation honestly and frame the trade-off in a way that supports clinical decision-making.

WHY THE ANSWER WORKED:

- Opened with the commitment to directness: "I'm going to give you the numbers straight, because you're asking exactly the right question." This signals

you're not hiding anything.

- Named the bleeding increase upfront: 3.8% vs. 2.1%, HR 1.82. Gave the absolute increase clearly: ~1.8 percentage points.
- Immediately contextualized against the MACCE benefit: 4 percentage points MACCE reduction vs. 1.8 percentage points bleeding increase. This is the trade Patricia needs to see quantified.
- Distinguished GI bleeding from ICH: This is critical. GI bleeds are manageable, reversible, and rarely fatal. ICH is catastrophic. Showing that ICH rates were balanced (0.4% vs. 0.3%) is a major safety reassurance.
- Closed with the benefit-risk framing P&T can use: "2 MACCE events prevented for every 1 major bleeding event caused." This is exactly how you'd structure a P&T slide.

KEY CLINICAL CONCEPT — TIMI Major Bleeding:

TIMI (Thrombolysis in Myocardial Infarction) major bleeding is a standardized scale used in cardiovascular trials to define clinically significant bleeding events:

- Intracranial hemorrhage (any)
- Clinically overt bleeding with hemoglobin drop ≥ 5 g/dL
- Fatal bleeding

Most TIMI major bleeding events in antiplatelet trials are GI bleeds (upper or lower GI tract) — these are serious but typically manageable with endoscopy, transfusion, and temporary drug interruption.

KEY CLINICAL CONCEPT — Intracranial Hemorrhage (ICH) vs. GI Bleeding:

ICH (intracranial hemorrhage) — bleeding into the brain — is the catastrophic complication of antiplatelet therapy. It carries high mortality (30–50%) and permanent disability in survivors. GI bleeding — though classified as "major" by TIMI criteria — is far less likely to be fatal. Most GI bleeds are identified early, treated with endoscopy or surgery, and patients recover fully after drug discontinuation. The fact that ICH rates were balanced (0.4% vs. 0.3%) is a critical

safety finding: the excess bleeding risk is driven by manageable GI events, not catastrophic brain bleeds.

KEY CLINICAL CONCEPT — Benefit-Risk Ratio:

For every 100 patients treated for 3 years with cardiovexaban:

- **4 MACCE events prevented** (CV death, MI, stroke, urgent revascularization)
- **2 major bleeding events caused** (mostly GI bleeds)

Net clinical benefit: 2 more favorable outcomes than adverse outcomes per 100 patients treated. This ratio is considered acceptable in high-risk secondary prevention populations, where the baseline MACCE risk is high and the absolute benefit outweighs the bleeding risk.

⚠ **WHAT TO WATCH FOR:** Never present efficacy data without proactively addressing bleeding risk. If Patricia has to ask about bleeding after you've spent five minutes on MACCE reduction, you've failed the credibility test. Lead with benefit-risk balance, not just benefit.

Closing Challenge — The Cost Question: Not Yet Answered

Dr. Holloway sets down her pen and looks at you for a long moment, then gives a small nod.

PATRICIA:

"That's a good answer. You didn't try to bury the bleeding risk, and you gave me the framing I need — 2 MACCE prevented for every 1 bleed caused. I can work with that. My cardiologists will buy that trade in the right patient population."

She flips to a new page on her tablet.

PATRICIA:

"The ICH data is reassuring — that's the outcome I worry about most with dual-pathway inhibition. GI bleeds I can manage. Brain bleeds I can't."

She looks up.

PATRICIA:

"Which brings me to the question I haven't asked yet, but you know is coming. Let's talk about cost. I've got generic clopidogrel at pennies per day. I've got ticagrelor on contract at a reasonable rate. Cardiovox is going to come in at a premium — how much of a premium, and how do you justify that incremental cost given the NNT we discussed?"

She's watching your face as she asks, clearly testing whether you'll handle this one as directly as you handled bleeding risk.

TRAINEE CHALLENGE — Your Turn

This Is Where the Simulation Ends — and Your Preparation Begins.

Patricia just asked the cost-effectiveness question. The conversation is now in your hands. Before you read the annotated answer in a future session, stop here and answer these questions yourself — out loud, in writing, or with a colleague playing Dr. Holloway.

Five Questions to Answer Before Your Next Session:

1. **What is the annual cost of cardiovascular compared to generic clopidogrel and branded ticagrelor?** What's a realistic WAC (Wholesale Acquisition Cost) for a novel antiplatelet agent, and how much lower would the net cost be after rebates?
2. **What is the cost per MACCE prevented?** Given an NNT of ~35 over 3 years, and an annual drug cost (hypothetically) in that range, what's the incremental cost per event prevented compared to standard therapy?
3. **How do you frame the value proposition for a P&T committee?** What are the downstream cost offsets from preventing MACCEs (hospitalization costs, rehab costs, long-term disability costs)?
4. **What's your response to "generic clopidogrel costs pennies per day"?** How do you acknowledge that price difference without being defensive, and pivot to clinical differentiation?
5. **Patricia framed this as a cost-justification challenge — what does that signal?** What is she testing by asking this question last, after you've

established clinical credibility?

The annotated answer to Exchange 6 — including the cost-effectiveness framing, budget impact modeling considerations, and the full P&T value argument — will be provided in the Level 1 Continuation module. Come to that session prepared.

Level 1 Debrief — What This Simulation Tests

This simulation was designed to evaluate foundational clinical proficiency — the baseline floor of knowledge required to have a productive conversation with a P&T pharmacist. Below is a summary of what was tested and what the scoring distinction looks like.

What Was Evaluated

- **Patient selection precision:** Do you know who qualifies and who doesn't? Can you say "not for them" without hedging?
- **Risk stratification literacy:** Can you explain the ASCVD threshold, its design rationale, its variability, and its edge cases — without speculating beyond the data?
- **Data fluency:** Can you cite efficacy numbers (hazard ratios, CIs, p-values, NNT) and immediately translate them into clinical and economic language?
- **Honest handling of limitations:** ASCVD threshold limitations, CV death subgroup caveats, bleeding risk absolute increases — did you surface these proactively or wait to be caught?
- **Benefit-risk balance:** Did you present bleeding data with the same directness as efficacy data, and quantify the trade-off in terms P&T can use?
- **Conversational intelligence:** Pacing, pivoting, not over-explaining when a crisp answer is what's needed.

What Separates a Level 1 Pass from a Level 2 Challenge

Patricia in this simulation was engaged but not adversarial. She signaled what she wanted. She appreciated honesty and rewarded depth. She gave you room to recover. That is what Level 1 looks like.

In **Level 2**, the physician will not signal what they want — they'll misquote data and see if you correct them. They'll conflate hazard ratio and percent reduction and wait. They'll ask about head-to-head comparisons that don't exist. They'll say "my cardiologist told me the ASCVD threshold was 10%." You'll have to hold the line without Patricia's helpful pen-tapping and appreciative nods.

In **Level 3**, they'll use your answer against you. A concession you make in exchange 2 will reappear in exchange 5 as a setup. The conversation will feel collaborative until it isn't.

"She wasn't grading your product knowledge. She was deciding whether you're the kind of rep she returns calls for."

— Level 1 Simulation Core Principle

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